SOP, POS, Minterms & Maxterms

1. Sum of Products (SOP) & Product of Sums (POS)

Theory:
Boolean expressions are generally written in one of two formats:

1. **SOP (Sum of Products):** Two or more ANDed terms that are ORed together.

- *Example:* $F = xyz + xy$

2. **POS (Product of Sums):** Two or more ORed terms that are ANDed together.

- *Example:* $F = (A + B)(A + \bar{B}C)$

Theory:
A **Standard (or Canonical) Form** means that *every single term* in the expression must contain all the variables present in the domain.

- If the domain is A, B, C, D, then the term $A\bar{B}C$ is non-standard because it is missing D.

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2. Converting to Standard SOP

Theory:
To add a missing variable to an SOP term, you multiply the term by 1. In Boolean algebra, $1 = (X + \bar{X})$ where X is your missing variable.

(Example 1: Adding one missing variable D):

$$\begin{aligned} A\bar{B}C &= A\bar{B}C \cdot 1 \\ &= A\bar{B}C(D + \bar{D}) \\ &= A\bar{B}CD + A\bar{B}C\bar{D} \end{aligned}$$

(Example 2: Adding two missing variables C and D):

$$\begin{aligned} \bar{A}\bar{B} &= \bar{A}\bar{B} \cdot 1 \cdot 1 \\ &= \bar{A}\bar{B}(C + \bar{C})(D + \bar{D}) \\ &= \bar{A}\bar{B}(CD + C\bar{D} + \bar{C}D + \bar{C}\bar{D}) \\ &= \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} \end{aligned}$$

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3. Converting to Standard POS

Theory:
To add a missing variable to a POS term, you add 0 to the term. In Boolean algebra, $0 = X\bar{X}$ where X is your missing variable. Then, you apply the distributive law: $A + BC = (A + B)(A + C)$.

(Example 1: Expanding $F(A, B) = (\bar{A} + B)B$):

$$\begin{aligned} F(A,B) &= (\bar{A} + B) \cdot (B) \\ &= (\bar{A} + B) \cdot (B + 0) && \text{(Add 0 to the incomplete term)} \\ &= (\bar{A} + B) \cdot (B + A\bar{A}) && \text{(Replace 0 with } A\bar{A}\text{)} \\ &= (\bar{A} + B) \cdot (B + A)(B + \bar{A}) && \text{(Apply distributive law)} \end{aligned}$$

(Example 2: Expanding $F(A, B, C) = (A + B)(B + \bar{C})(\bar{A} + C)$):

$$\begin{aligned} &= (A + B + 0) \cdot (B + \bar{C} + 0) \cdot (\bar{A} + C + 0) \\ &= (A + B + C\bar{C}) \cdot (A\bar{A} + B + \bar{C}) \cdot (\bar{A} + B\bar{B} + C) \\ &= (A + B + C)(A + B + \bar{C}) \cdot (A + B + \bar{C})(\bar{A} + B + \bar{C}) \cdot (\bar{A} + B + C)(\bar{A} + \bar{B} + C) \end{aligned}$$

(Note: In POS, duplicate terms like $(A + B + \bar{C})$ can be reduced to a single term using the rule $X \cdot X = X$.)

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4. Minterms and Maxterms

Theory:

- **Minterms (m):** Used in SOP. The variable is normal if it is 1, and complemented (bar) if it is 0.
- **Maxterms ():** Used in POS. This is the exact opposite logic. The variable is normal if it is 0, and complemented (bar) if it is 1.

(2-Variable Minterm & Maxterm Truth Table):

Decimal	A	B	Minterm (SO)	Designation	Maxterm (OS)	Designation
0	0	0	$\bar{A}\bar{B}$	m_0	$A + B$	$_0$
1	0	1	$\bar{A}B$	m_1	$A + \bar{B}$	$_1$
2	1	0	$A\bar{B}$	m_2	$\bar{A} + B$	$_2$
3	1	1	AB	m_3	$\bar{A} + \bar{B}$	$_3$

Theory:
We can express Boolean functions as a sum of specific minterms using the Σ symbol.

(Example):

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\begin{aligned}
F(A,B) &= \Sigma m(1,2) \\
&= m_1 + m_2 \\
&= \bar{A}B + A\bar{B} \quad \text{(his is the XOR logic)}
\end{aligned}
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